

Google Energy

Federal Energy Regulatory Committee has approved Google's request to buy and sell electricity to its newest subsidiary named Google Energy

New smoke alarm

A fire extinguisher company and a bioventure in Tokyo have launched a smoke alarm for deaf people that emits a pungent smell when smoke is detected

red. Vertical viewing angles are poor, horizontal angles are a bit better but loss of colour fidelity happens when viewing from any incline. There is lots of noise in dimmer scenes as well as a problem with banding in lighter intensities of blue. The BenQ G2420HD proved that looks aren't an indicator of performance, as it belied the cheap build to post some impressive scores.

The BenQ V2400 Eco had loads of issues — slight loss of darker shades, with banding. Reds and browns appear oversaturated. Orange looks a little too bright and artificially radiant. Highlighting is good, but due to oversaturation of reds there is loss of fine detail. In the reverse text tests, light green on grey looks blurry. In Display Mate, the reds blend in higher intensities. There is also an issue with pure green, where the gamut is not as wide, causing some banding. Red is also overly bright with very little detail and very little variation in shade intensity. Both Viewsonic monitors performed very similarly, with the VX2433wm showing decent colour rendition for a TN panel but the reds look artificially bright. In movies, there is a noticeable lack of contrast and darker shades tend to blend. Banding is noticeable in such cases

HOW WE TESTED

Monitors were divided on the basis of screen size. On the features front we logged all the physical specifications including connectivity and extras if any. We also rated the build quality, menu system and bezel thickness of each panel, after all, usability and aesthetics are important. The stand mechanism is also graded.

Our test system consisted of an Intel Core 2 QX9650 CPU, ASUS P5Q3 Deluxe motherboard, ASUS Radeon EAH 5850 graphics card, 3 GB of DDR3 RAM, and a Western Digital 10,000 rpm Velociraptor hard drive. We used our Spyder 3 Elite to calibrate each display and also give us the actual contrast ratio and luminance figures. For measuring maximum luminance we kept the brightness at 100 per cent for all displays. For getting a regular luminance value and for all other Spyder 3 Elite scores we set the brightness and contrast at 50 per cent each. Luminance of white and black was also logged. The higher the white point the better — for this means a display will do really bright colours and highlighting and HDR will look more realistic. The lower the black point i.e. closer to zero,

and in general wherever there is a lot of a particular shade in varying intensities. The VG2427wm had issues in the reverse text tests, where dark yellow on dark grey is not very visible. Bright yellow on white is also barely legible. When a dark image is viewed from a slight angle there is purple/copper hue. Smudging and banding occurs in darker scenes in movies. Games were decent on these monitors and the pixel response is good. The ASUS VH242

let down in games — surprising, really, for we felt it was aimed at that fraternity. There was an issue with a delay in backlight adjustment. This spoils Crysis, as it was noticeable if you are turning away from facing a bright spot, to face a dim region.

the better is the ability of the display to produce deepest blacks. Therefore a black luminance of 0.12 is better than 0.20 or 0.15.

Post calibration we ran Display Mate to judge the contrast ratio for ourselves via the grey scale intensity test which consists of checking visibility of grey squares of varying intensity on a black background. Display Mates colour test suite allowed us to check for colour gamut and shade intensity. In the colour gamut test we test for the wideness of the colour gamut where the primary colours are separated by the secondary colours and the entire spectrum is presented on-screen. Colours should smoothly merge into others without vertical lines called "banding". When testing for purity of the primary colours and individual shades we use the 16-shade ramp test and the 256-ramp test. 16 and 256 bands (respectively) are presented for each of the primary and secondary colours one colour at a time and these bands vary equally in intensity moving from highest intensities to lowest intensities. We check for uniformity of change in intensity

and adequate separation between each band — this tells us that the monitor is able to display a variable intensities of a particular shade well. Generally IPS and PVA panels that are natively 8-bit are better performing than 6-bit TN panels that use dithering to make up for additional colour intensities that they cannot produce.

Our movie test consisted of a set of four HD 1080p clips encoded in WMV format. A lot of scenery which allows for checking colour and contrast, fast action scenes which allow us to test the pixel response. There is a volcano eruption scene after dark, this has a lot of grey on black as smoke rises and this allows us to check for detail in darker scenes i.e. contrast and blackness level. Underwater scenes allow for testing per pixel detail banding and a scene shot in caves allow us to check for colour accuracy and detail in darker areas. Crysis was used for game tests and we patched it to the latest version. We looked for detail in darker areas, the colour of scenery, highlighting and HDR and pixel detail as well as tonal accuracy of colours.

At 50 per cent brightness and contrast, this display is also dimmer than other monitors. Yellow on white and cyan on green are colour registration problem areas for reverse text. The gamut for green is wide though not so for blue and red.

And the award goes to

The BenQ M2400HD and G2420HD were very close to each other in all the tests but the M2400HD has a lot more features that puts it ahead. But the pricing brings them closer in terms of overall value and in this regard they're inseparable — a joint Best Buy award for BenQ. The G2420HD also wins our Editor's Pick award owing to its great pricing of Rs. 12,500 — superb value. The Alienware

BEST BUY
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BenQ M2400HD

BEST PERFORMER
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Alienware OptX AW2310

OptX AW2310 wins our Best Performer award owing to its great all round performance.

22-inches: Jacks-of-all-trades?

22-inches is the magical category at the moment, the one that is moving volumes, making profits and also subject to the most competitive tactics as brands try to outthink each other and guess what users